

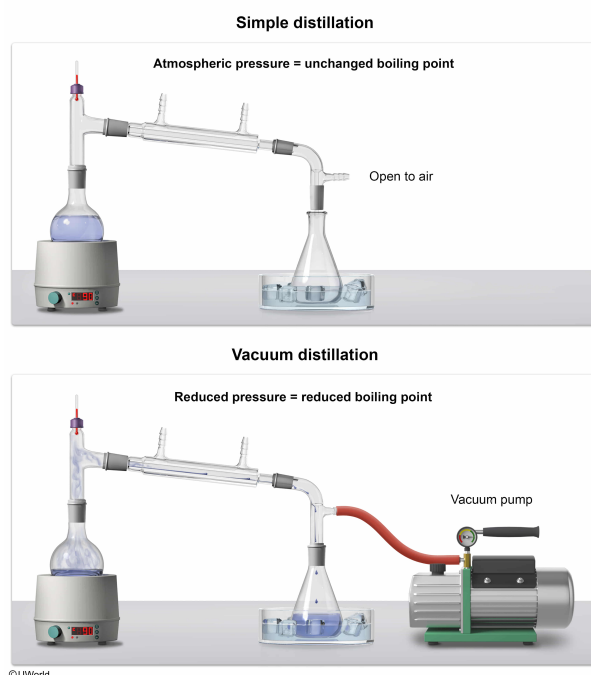
Researchers purify two samples of the same mixture, one by simple distillation and the other by vacuum distillation. Which of the following expressions correctly compares the relationship between the mixture components' boiling points (bp) during each distillation?

- ☐ A. Simple distillation bp = vacuum distillation bp [4%]
- ☐ B. Simple distillation bp < vacuum distillation bp [45%]
- ✓ ☒ C. Simple distillation bp > vacuum distillation bp [49%]
- ☐ D. Simple distillation bp = 2(vacuum distillation bp)

Correct

49% answered correctly

Explanation:



Distillation is a technique used to **separate** liquid molecules based on their **boiling point**, the temperature at which the vapor pressure equals the pressure of the system. There are three common types of distillation: **simple**, **fractional**, and **vacuum**. The type of distillation suitable for a mixture is dependent on the boiling points of the mixture components.

Simple distillation is used for compounds with boiling points less than 150 °C, and that are more than 25 °C apart from each other. Vacuum distillation is used for compounds that decompose at their boiling point (typically boiling points greater than 150 °C), causing the boiling point to decrease and thereby preventing degradation. Simple and vacuum distillations are set up in the same way, except that simple distillations are done at **atmospheric pressure**, whereas **vacuum distillations** are connected to a vacuum pump and performed at **reduced pressure**.

Because vacuum distillations are performed under reduced pressure, the boiling point of a compound under vacuum will **decrease** relative to its **boiling point** at atmospheric pressure. Therefore, the simple distillation boiling point of a molecule is greater than the boiling point of that molecule under vacuum (ie, **simple distillation bp > vacuum distillation bp**).

(Choices A, B, and D) Reducing the pressure of a system reduces the boiling point of a compound. Therefore, the boiling points in simple and vacuum distillations cannot be equal and the vacuum distillation boiling point would not be greater than the simple distillation boiling point.

Educational objective:

Distillation is a technique used to separate molecules based on their boiling points. The common types of distillation are simple, fractional, and vacuum. Simple distillations are done at atmospheric pressure, whereas vacuum distillations are done at a reduced pressure, decreasing the compound's boiling point relative to the boiling point at atmospheric pressure.

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System:
Separation Techniques, Spectroscopy, and Analytical
Methods